



Semester 1 Examinations 2018 / 2019

Exam Code(s)	1MDM1, 1MF1, 1SD1, 1SD3
Exam(s)	Masters in Software Design & Development Higher Diploma in Applied Science (Software Design & Development) Masters in Digital Media
Module Code(s)	CT870
Module(s)	Internet Programming
Paper No.	1
External Examiner(s)	Prof. Pier Luca Lanzi
Internal Examiner(s)	Prof. Michael Madden *Dr. Enda Barrett

Instructions: Answer any 3 questions. All questions carry equal marks.

Duration	2hrs
No. of Pages	3 (Including cover page)
Department(s)	Information Technology

Requirements None

Question 1:

- a) Explain the steps involved in a browser requesting and displaying a simple HTML webpage. (4 marks)
- b) Describe 6 different tags that can be used on a HTML page. Illustrate each one with an example. (4 marks)
- c) Write HTML code that uses a table to display a single special offer product for a fictional e-commerce site. You may choose any product you like. A header cell should be used to contain the title, description, price, and image (the src attribute can contain a mocked link) of the product. Standard cells should be used to display the data about the single item instance. Using the colspan attribute ensure that the description spans a width of two columns. (6 marks)
- d) Write HTML for a login form which contains a dropdown list (options for Male/Female), two text boxes (one for username and one for password), a checkbox for “remember me”, a submit and a reset button. (6 marks)

20 Marks

Question 2:

- a) Explain how the grid system in Bootstrap works. In your answer discuss the purpose of rows and columns and describe how the grid facilitates responsive design. (5 marks)
- b) Provide the HTML and corresponding Bootstrap classes to split a row into two separate sections, each sized 6 columns wide on a large device and medium device. In the case of smaller devices, the columns should stack on top of each other. (5 marks)
- c) What are style classes? Give an example of a class and an id based CSS selector. (5 marks)
- d) In CSS describe using an example how a single style can be applied to all tags of a particular type on the web page. (5 marks)

20 Marks

Question 3:

- a) Using JavaScript, write code to iterate through an array of objects (called myArray), removing objects placed at **odd** index positions in the array. Assume the array contains 8 vehicle objects which take the following form ({**make**: "Ford", **model**: "Focus", **age**:2}, ... , {**make**: "Toyota", **model**: "Yaris", **age**:5}). You may use the splice method to remove the objects from the array, however each time splice is called it will re-index the array. To counter this you should iterate through the array in reverse. (9 marks)
- b) Whilst the splice method can be used to remove items from an array, please list a method which is commonly used to add items to an array (3 marks)
- c) Write JavaScript code that upon button click reads 3 numbers from 3 text boxes and doubles each number before saving it in the text box it came from. (8 marks)

20 Marks

Question 4:

You have been asked to build a NodeJS (Express) backend, which comprises a collection of RESTful APIs for an Instagram style application. Each image shared by the users of the application should contain a description of the image, the upload time and date, the number of likes and the number of re-shares.

- a) Provide a sample JSON document which contains data for a single image shared on the platform. (4 marks)
- b) Write either a MongoDB or Mongoose query (whichever you prefer) which returns all images posted, sorted in **ascending** order by upload time. (5 marks)
- c) Define a server-side API route accessible at the path '/uploadImages' which accepts POST requests. Assume that request body contains JSON data and the payload is processed by the Express middleware and made available as an object in the request body. Either using the object modelling tool Mongoose or a traditional MongoDB query, provide code to save the data in the request body to the MongoDB database. If using Mongoose, you do not need to specify the schema, you may assume a schema named Image is already defined. (7 marks)
- d) Define a server-side API to delete documents from the collection by parsing the request parameters, extracting the id and removing the document found. (4 marks)

20 Marks